Machine Learning Models Explained

Three-page summary of the algorithms, their purpose, and the model-selection approach used in the project.

REGRESSION	CLASSIFICATION	CLUSTERING
Estimate delivery minutes	Estimate late-order risk	Discover similar groups

01 Project Objective & Model Map

The project applies supervised and unsupervised learning to delivery operations. **Regression** predicts delivery duration in minutes; **classification** predicts whether an order will exceed the selected service-level threshold (SLA); and **clustering** groups delivery locations and rider records with similar observed characteristics.

Business question	ML task	Algorithms	Output
How long may delivery take?	Regression	Ridge; Random Forest; Mean baseline	Minutes
Will the order be late?	Binary classification	Logistic Regression; Random Forest	Late probability / class
Where is demand concentrated?	Clustering	K-Means	Demand-zone groups
Which riders have similar profiles?	Clustering	K-Means	Rider segments

02 Delivery-Time Prediction — Regression

Regression predicts a continuous numerical target. In this project, the target is **Time_taken(min)**, the time taken to complete a delivery.

Ridge Regression

Definition: A linear regression model that adds an L2 penalty to large coefficients. This regularisation helps control overfitting, especially when input variables are correlated.

- **Why used:** It gives a relatively simple, interpretable linear reference model and helps test whether permitted features have broadly linear relationships with delivery time.
- **Limitation:** It may not capture complex nonlinear patterns or interactions.

Random Forest Regressor

Definition: An ensemble of decision trees trained on random samples of the training data. The final numerical prediction is the average of the trees' predictions.

- **Why used:** Delivery time can depend on nonlinear combinations of distance, traffic, weather and order timing. Random Forest can learn these patterns without assuming a straight-line relationship.
- **Strength / limitation:** It captures interactions and provides feature-importance estimates, but is more complex to explain than Ridge Regression.

Mean Dummy Regressor — baseline

Predicts the average delivery time calculated from the training set for every order. It does not learn feature relationships. We use it as a simple benchmark: a useful ML model should improve on this baseline on unseen data.

Regression model selection: Compare Ridge and Random Forest using validation MAE. Choose the candidate with the lowest validation MAE, then evaluate it once on the held-out test set. The winning model must be confirmed from the executed notebook results.

03 Late-Order Prediction — Classification

Classification predicts a category rather than a continuous number. The target is **is_late**: 1 indicates delivery time exceeds the selected SLA; 0 indicates it does not. The 30-minute threshold in the current EDA is illustrative unless the challenge organiser confirms it.

Logistic Regression

Definition: A classification algorithm that estimates the probability that an order belongs to the late class. The probability is converted to a class using a decision threshold.

- **Why used:** It is a relatively simple, interpretable classification baseline. Its coefficients help describe how features are associated with late-order risk.
- **Limitation:** Standard Logistic Regression may not capture complex nonlinear relationships.

Random Forest Classifier

Definition: An ensemble of decision trees that combines their votes or class probabilities to classify an order as late or on time.

- **Why used:** It can learn nonlinear patterns and interactions among distance, traffic, weather and order timing.
- **Limitation:** The combined decisions are less straightforward to interpret than a small individual decision tree.

How classification models are selected

Candidate models are compared using **validation PR-AUC**. The classification threshold is also chosen using validation data and the project's relative-cost assumptions for missed late orders versus unnecessary interventions. The test set is kept for final evaluation only.

Metric	What it tells us	Desired direction / use
Precision	Of the orders flagged late, how many are actually late?	Higher when false alerts are costly
Recall	Of all actual late orders, how many did the model catch?	Higher when missed delays are costly
F1-score	Balances precision and recall.	Higher
PR-AUC	Summarises precision-recall performance across thresholds.	Higher; useful with class imbalance
ROC-AUC	Measures ranking ability across thresholds.	Higher

04 Model Evaluation & Selection Workflow

Data stage	Role in this project	Important rule
Training set	Fit preprocessing steps and candidate models.	Learn only from training data.
Validation set	Compare candidates and choose model / threshold.	Use for selection, not final reporting.
Test set	Measure final performance of selected model.	Do not tune or select using test results.

Regression metrics

Metric	Definition	Better value
MAE	Average absolute prediction error, in minutes.	Lower
RMSE	Penalises larger prediction errors more strongly.	Lower
R^2	Measures the share of target variation explained by the model.	Higher

Why compare different algorithms? Ridge and Logistic Regression provide simpler, interpretable reference models. Random Forest models can capture nonlinear relationships. The baseline verifies that the regression models add value beyond predicting a constant average.

05 Unsupervised Learning — K-Means Clustering

Clustering groups similar observations when predefined target labels are unavailable. **K-Means** assigns observations to K groups so that points within each group are relatively similar to one another based on the selected features.

Demand-zone clustering

K-Means groups customer delivery destinations using validated geographical coordinates and justified demand information. The purpose is to highlight areas with concentrated delivery demand that may deserve further investigation as potential service or dark-store zones.

Rider performance segmentation

K-Means groups riders using aggregated measures such as average delivery time, late rate, rating and delivery volume, where available and reliable. The resulting clusters describe observed performance profiles; they do not explain the causes of individual performance.

Choosing and interpreting clusters

- Use elbow diagnostics to examine how within-cluster variation changes as K increases.
- Use silhouette score to assess how well-separated and internally consistent the clusters are.
- Inspect cluster profiles before assigning descriptive labels. A cluster centroid is not automatically an optimal dark-store location.

06 Why This Model Combination?

Task	Selected algorithms	Reason for the combination
Delivery-time prediction	Ridge; Random Forest Regressor; mean baseline	Linear reference + nonlinear ensemble + simple benchmark.
Late-order prediction	Logistic Regression; Random Forest Classifier	Interpretable probability model + nonlinear classifier.
Demand-zone discovery	K-Means	Finds geographical groups without predefined labels.
Rider segmentation	K-Means	Finds groups with similar observed delivery profiles.

07 Modelling Safeguards & Limitations

- **Prevent data leakage:** exclude pickup-time and pickup-delay information from predictions intended to be made when an order is placed.
- **Pipeline preprocessing:** fit imputers, encoders and scalers within training pipelines; do not learn preprocessing statistics from validation or test data.
- **Fair evaluation:** use validation data for model selection and reserve the test set for final reporting.
- **Interpretation:** EDA patterns and feature importance show associations, not proof of causation.
- **Dataset scope:** the supplied data is a food-delivery dataset used as a quick-commerce proxy; it does not provide verified dark-store operations or every real-world site-selection factor.

Final takeaway
The project uses regression to estimate delivery minutes, classification to flag late-order risk, and K-Means to discover demand and rider groups. Regression is selected by the lowest validation MAE; classification is selected by validation PR-AUC with a validation-based decision threshold. The actual winning models and final scores should be taken from the executed notebook comparison outputs.